CLINICAL · EAR

The dizzy child

A dizzy child is rarely a sick child, but the history is harder to get and the causes are different from an adult's. Migraine in its childhood forms, glue ear and a viral vestibular upset account for most cases; true BPPV and Ménière's disease are rare. Your job is to sort the timing of the attacks, check hearing and the ears, look hard for the few signs that mean the brain or the inner ear's structure is at fault, and reassure the rest with confidence.

AT A GLANCE

- BPVC: seconds-long spells at 2 to 6 years, normal between, no hearing loss; a migraine precursor.
- Vestibular migraine leads in older children; family history in most.
- Glue ear makes children clumsy rather than giddy.
- BPPV in a child almost always follows head injury.
- Red flags: ataxia between attacks, morning headache, focal signs, central nystagmus.

1 CHAPTER ONE Overview

What a dizzy child is telling you.

1.1 Why children are different

Young children cannot describe vertigo. They cling to a parent, fall, refuse to walk, go pale, vomit, or stop playing, and the parent reports "clumsiness" or "falls a lot". The examination is limited by cooperation and has to be built from play and observation. Children also compensate quickly after a vestibular loss, so a once-dramatic attack may leave few signs a week later.

1.2 How common it is

Dizziness is an uncommon referral in children and most causes are benign. Over half of dizzy children also have headache, and a first-degree family history of migraine is present in most children whose vertigo turns out to be a migraine variant.

COMMON AND IMPORTANT CAUSES BY TYPE OF ATTACK

PATTERN	LEADING CAUSES	CLUES
Brief spells of seconds	Benign paroxysmal vertigo of childhood; BPPV after head injury	Toddler, no hearing loss, normal between attacks; positional after trauma
Attacks of minutes to hours	Vestibular migraine; rarely Ménière's disease	Headache, light and sound sensitivity, family history; fluctuating hearing and fullness
A single attack lasting days	Vestibular neuritis; labyrinthitis	Viral prodrome; hearing normal in neuritis, reduced in labyrinthitis
Unsteadiness rather than spinning	Otitis media with effusion	Conductive hearing loss, flat tympanogram
Life-long imbalance, late walking	Congenital or acquired severe vestibular loss	Sensorineural hearing loss, meningitis, cytomegalovirus, inner ear malformation
Spells with fatigue and screens	Refractive error or convergence problems	End of the day, headache, rarely vomiting
Feeling faint on standing	Orthostatic hypotension	Rapid growth around puberty; presyncope, not spinning

CHAPTER TWO

2 Causes

From the common to the dangerous.

2.1 Benign paroxysmal vertigo of childhood

The commonest cause between 2 and 6 years, accounting for up to a third of dizzy children in that age group. A *migraine precursor*: sudden, intense spells of vertigo lasting seconds to minutes, often with pallor, fear, nystagmus and unsteadiness, sometimes nausea or vomiting, without a trigger and *without* hearing symptoms. The child is entirely normal between spells. About half go on to have migraine in adolescence, and most have a family history of migraine. The outlook is excellent and it resolves.

OTHER CHILDHOOD MIGRAINE VARIANTS

Benign paroxysmal torticollis (episodic head tilt in infants under 2), *cyclical vomiting* and *motion sickness* (most marked between 4 and 10 years) belong to the same family and often precede vestibular migraine.

2.2 Vestibular migraine

The commonest cause in older children and adolescents: attacks of rotatory vertigo lasting minutes to hours, accompanied or followed by headache and often vomiting, with sensitivity to light and sound. A family history of migraine is present in most. It must be told apart from *basilar-type migraine*, in which vertigo is part of an aura with paraesthesia, ataxia or loss of consciousness, because the treatment differs: ergot preparations are contraindicated in basilar migraine.

2.3 Otitis media with effusion

Probably the commonest cause of balance complaints in young children overall. The child is unsteady and clumsy rather than spinning, with a conductive hearing loss. Proposed mechanisms are pressure changes at the oval and round windows and toxins from the middle ear fluid crossing into the inner ear. Balance often improves once the effusion clears or grommets are inserted.

2.4 Vestibular neuritis and labyrinthitis

As in adults: acute severe rotatory vertigo for days, with nausea, vomiting, imbalance and oscillopsia, often after a viral illness. Hearing is normal in neuritis and reduced in labyrinthitis. Children recover faster than adults because they compensate well, but they are often admitted and a central cause is considered first.

2.5 Chronic ear disease

Cholesteatoma can erode the lateral semicircular canal and cause a fistula, with vertigo on pressure (a positive fistula test). Suppurative labyrinthitis is a

complication of acute or chronic otitis media and leaves a dead ear.

2.6 Inner ear malformations and syndromes

Vestibular abnormalities accompany up to 70% of permanent sensorineural hearing losses in children. These children present with late motor milestones and imbalance rather than vertigo. Some malformations, especially the *enlarged vestibular aqueduct* and the incomplete partition anomalies, cause episodic vertigo with stepwise hearing loss after minor head injury, and are easily mistaken for Ménière's disease. Syndromic associations include Pendred, branchio-oto-renal and Waardenburg syndromes, and *Usher syndrome*, in which deafness and absent vestibular function are joined later by retinitis pigmentosa. A profoundly deaf child with absent vestibular function needs an eye examination with electroretinography. Meningitis and congenital cytomegalovirus can destroy both cochlear and vestibular function.

2.7 Head injury

A temporal bone fracture through the labyrinth, or a perilymph fistula, causes vertigo with tinnitus and hearing loss. *Post-concussion syndrome* causes dizziness without hearing loss, and *BPPV* in children is almost always post-traumatic or after ear surgery such as cochlear implantation.

2.8 Third-window disorders

Superior canal dehiscence is rare in children but presents as in adults, with sound-induced vertigo (*Tullio phenomenon*), pressure-induced vertigo (*Hennebert sign*) and chronic disequilibrium.

2.9 Drugs

Aminoglycosides, including gentamicin drops into a perforated ear, are vestibulotoxic and cause ataxia and oscillopsia, often without hearing loss. Cisplatin and antimalarials are mainly cochleotoxic.

2.10 Vision

Refractive errors and convergence problems cause brief vertigo in about 10% of dizzy children over 5, usually at the end of the day or after long screen time, often with headache and nausea. An optometry review is part of the work-up.

2.11 Psychological causes

Anxiety and depression can present as dizziness, and long-standing vestibular symptoms can produce them. Prolonged school avoidance is a marker. *Medically unexplained dizziness* is a diagnosis of exclusion, and most children improve with therapy-based treatment.

2.12 Central causes not to miss

Posterior fossa tumours, Chiari malformation, hereditary ataxias, encephalitis, multiple sclerosis in adolescents, and *epilepsy*: temporal or occipital lobe seizures can produce a sensation of movement with visual or auditory hallucinations, and a prolonged recovery may be a post-ictal state. Cardiac arrhythmias cause presyncope rather than vertigo.

RED FLAGS IN THE DIZZY CHILD

- Persistent ataxia or unsteadiness between attacks.
- Headache on waking or with vomiting, or a headache that changes in character.
- Focal neurological signs, abnormal eye movements, or nystagmus that is vertical or changes direction with gaze.
- Loss of consciousness, aura with hallucinations, or confusion afterwards.
- Progressive or asymmetric sensorineural hearing loss.
- Regression of motor milestones.

Any of these calls for MRI of the brain and internal auditory meati and paediatric neurology input.

3 CHAPTER THREE Assessment

History first, then play.

3.1 History

Take the history from the child and the parent, and try to make the child describe it ("does the room spin, or do you feel wobbly inside?").

- **Timing:** how long each spell lasts, how often, and whether there was a single attack or many.
- **Triggers:** movement, position change, rolling or looking up, straining, loud noise, standing up, screens, tiredness, missed meals.
- **Hearing:** loss, fluctuation, tinnitus, fullness, ear discharge, previous grommets or ear surgery.
- **Headache:** before, during or after, with light or sound sensitivity; motion sickness; family history of migraine.
- **Milestones:** age of sitting (about 6 to 9 months) and walking (about 12 to 15 months), and balance tasks such as riding a bicycle without stabilisers. Delay suggests a congenital or early vestibular loss.
- **Past history:** prematurity, meningitis, cytomegalovirus, ototoxic drugs, head injury, surgery.
- **Vision, school and mood.**

3.2 Examination

Grade the examination to the child's cooperation and get the most useful information first.

- **Ears:** for effusion, perforation, cholesteatoma; pneumatic otoscopy for a fistula sign.
- **Eyes:** spontaneous and gaze-evoked nystagmus, ideally with Frenzel lenses or video goggles to remove fixation; avoid extreme gaze, where physiological nystagmus appears.
- **Head impulse test:** the child fixes on a cartoon picture while the head is turned quickly through about 15 degrees; a corrective saccade marks the weak side.

- **Head-shake test:** eyes closed, head tilted down 30 degrees and shaken 20 times; nystagmus afterwards indicates asymmetry.
- **Standing on one foot:** briefly at 30 months, 2 seconds at 3 years, 5 seconds at 4 years and 10 seconds at 5 years.
- **Romberg and Unterberger tests,** gait, and a cerebellar and cranial nerve examination.
- **Dix-Hallpike test** only if the history suggests positional vertigo, usually after head injury.
- **Blood pressure lying and standing** if the spells are presyncopal.

PERIPHERAL OR CENTRAL?

	PERIPHERAL	CENTRAL
Nystagmus	Horizontal-torsional, one direction, suppressed by fixation	Vertical or pure torsional, changes direction with gaze, not suppressed
Head impulse	Abnormal on the affected side	Usually normal
Balance	Falls to one side but can stand	Often cannot stand or walk
Other signs	Hearing loss may accompany it	Ataxia, cranial nerve or long tract signs, headache

3.3 Investigations

- **Audiometry and tympanometry** in every child: play audiometry, pure tones with reflexes, or auditory brainstem response depending on age. This single test separates the glue ear, the sensorineural loss and the normal-hearing groups.
- **Vestibular tests** where available: video-oculography with calorics, video head impulse testing, rotational chair testing (in infants on a parent's lap), cervical VEMPs adapted to children, and posturography from about 3 years. They show whether there is a unilateral or bilateral loss and separate vestibular from visual and proprioceptive causes.
- **MRI** of the brain and internal auditory meati when there are red flags, an unexplained or asymmetric sensorineural loss, or symptoms that do not settle; it may need a general anaesthetic in young children. **CT** of the temporal bones for chronic ear disease, trauma or suspected malformation.
- **Bloods** are rarely useful unless the history points to anaemia, diabetes or infection.
- **Optometry, ECG and EEG** guided by the history.

THE DIZZY CHILD

1 History

- → timing of spells
- → hearing
- → headache and family history
- → milestones
- → trauma and drugs

2 Examine

- → ears
- → nystagmus
- → head impulse and head shake
- → balance for age
- → neurology

3 Test

- → audiometry and tympanometry for all
- → vestibular tests where available

4 Red flags present

- → urgent MRI and paediatric neurology

5 Glue ear with imbalance

- → **treat the effusion**
  watchful waiting or grommets

6 **Normal ears and hearing**

→ **BPVC or vestibular migraine**
explanation, triggers, review

7 **Sensorineural loss**

→ aetiological work-up, MRI and CT, cytomegalovirus, genetics, eyes

4 CHAPTER FOUR Management

Explanation does most of the work.

4.1 Reassurance and explanation

Most families need a clear diagnosis and the knowledge that the cause is benign. Benign paroxysmal vertigo of childhood needs no treatment beyond explanation and review; it resolves, and parents should know that migraine may follow later.

4.2 Acute attacks

Vestibular sedatives can help briefly in a severe acute attack such as vestibular neuritis. Phenothiazines are used with caution in children because of extrapyramidal reactions, and never for long, since they slow compensation.

4.3 Vestibular migraine

Start with triggers: regular sleep, meals and fluids, limits on screen time, and attention to stress and vision. Drug prophylaxis, if needed, is best managed by a paediatrician or paediatric neurologist experienced in childhood migraine.

4.4 Otitis media with effusion

Manage as for any glue ear; imbalance strengthens the case for grommets if the

effusion persists.

4.5 Vestibular rehabilitation

Physiotherapy exercises speed compensation after vestibular neuritis, head injury or a fixed vestibular loss, and help children with chronic imbalance and the anxious, deconditioned child.

4.6 Specific conditions

BPPV, Ménière's disease, superior canal dehiscence and perilymph fistula are managed on the same principles as in adults. Cholesteatoma with a fistula needs surgery. Children with bilateral vestibular loss need early physiotherapy, safety advice about swimming and the dark, and occupational therapy input at school.

4.7 Referral

Refer to a tertiary paediatric audiovestibular or neuro-otology service when symptoms persist despite first-line management, when vestibular testing is not available locally, or when surgery is being considered.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Where the books differ

WHERE THE BOOKS DIFFER

Oxford Textbook
calls glue ear the commonest cause of balance complaints in children, BPVC the commonest cause of true vertigo between 2 and 6 years, and vestibular migraine the commonest cause in older children; it

stresses milestones, the one-foot standing test and the head impulse test.

Adult texts (Dhingra and K. J. Lee) mention childhood vertigo only briefly; their vestibular chapters apply to children for neuritis, BPPV and Ménière's disease, which behave as in adults.

5.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The causes **by duration of attack**, with BPVC, vestibular migraine and glue ear first.
- **BPVC** as a migraine precursor: seconds-long spells, normal between them, no hearing loss.
- Why every dizzy child needs **audiometry and tympanometry**.
- The **red flags** that need MRI and a neurologist.
- **Milestones** and the one-foot test as markers of vestibular loss.
- Vestibular loss with deafness: **enlarged vestibular aqueduct, Usher syndrome, cytomegalovirus, meningitis**.

5.3 Common pitfalls

COMMON PITFALLS

- Diagnosing BPPV in a child without a history of head injury.
- Missing glue ear because the child "is just clumsy".
- Calling enlarged vestibular aqueduct Ménière's disease.
- Forgetting the eyes, the blood pressure and the child's mood.
- Overlooking persistent ataxia between attacks.

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